Review notes · for you

Databases and SQL: unit check

Data engineering · Level 3 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Tables, keys and types
- Questions in SQL
- Joining tables
- Fast answers: indexes

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (15 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (8 min): Reveal each answer and ask one team to explain why it's right.

Discussion

- Which app's database would you like to see inside?
- When would a left join save you from a wrong answer?
- Which column would you index first, and why?

Check yourself

- One query I could now write that I couldn't before, and what it answers.

Try it this week
Sketch the tables of an app you use, write one JOIN you'd run on them, and say which column you'd index.

Quiz sheet

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Name _____ Class _____ Date _____

1 **Which column makes the best primary key for an app's users?**

◯ The user's name ◯ A user id given once and never reused ◯ The user's country

2 **A foreign key points to a row in another table.**

◯ True ◯ False

3 **Put the parts of a query in the order SQL needs them.**

☐ GROUP BY

☐ LIMIT

☐ SELECT

☐ ORDER BY

☐ WHERE

☐ FROM

4 **A table has 12 rows; one mark is NULL. What does this return?**

```
SELECT COUNT(*), COUNT(mark) FROM marks;
```

◯ 12 and 11 ◯ 11 and 11 ◯ 12 and 12

5 **Match each join to what it does with a member who never borrowed.**

JOIN (inner)	• • • •	Shows only members like them
LEFT JOIN members to loans	• • • •	Leaves them out
LEFT JOIN … WHERE book IS NULL	• • • •	Keeps them, with NULL for the book

6 **This should list members with no loans, but returns nothing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT name FROM members
2  LEFT JOIN loans
3  ON loans.member_id = members.id
4  WHERE book = NULL;
```

7 **After joining members to loans, which of these can happen? Choose all that apply.**

- [ ] A member with three loans appears three times
- [ ] COUNT(*) counts loans, not members
- [ ] Members are always counted once

8 **About how many looks does a binary search need to find one id among a million sorted ids?**

- ◯ Exactly 1,000
- ◯ About 500,000
- ◯ About 20

9 **Fill the gaps: count each class's loans with a dictionary.**

```
counts = {}
for c in ["5A", "6A", "5A"]:
    counts[c] = counts._____(c, _____) + 1
print(counts["5A"])
```

Word bank: `add` `get` `1` `0`

10 **Complete the sentence.**

An index makes searches _____, but it takes _____ and makes every new row slower to write.

Word bank: slower · faster · space · time off

Check your answers

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1 **A user id given once and never reused**

A primary key must be different in every row.

2 **True**

That's how tables link: photos.user_id points to a row in users, where id is the primary key.

3 **Top to bottom: 4, 6, 1, 5, 3, 2**

SELECT columns FROM a table WHERE rows match, GROUP BY a column, ORDER BY a column, LIMIT the number of rows.

4 **12 and 11**

COUNT(*) counts every row; COUNT(mark) skips the NULL.

5 **JOIN (inner) — Leaves them out**
LEFT JOIN members to loans — Keeps them, with NULL for the book
LEFT JOIN … WHERE book IS NULL — Shows only members like them

Inner joins drop rows with no match; left joins keep them as NULL, and IS NULL picks them out.

6 **Line 4:** `WHERE book IS NULL;`

Nothing equals NULL, so = NULL never matches.

7 **A member with three loans appears three times; COUNT(*) counts loans, not members**

A join makes one row per match, so COUNT(*) counts the pairs.

8 **About 20**

Each look halves what's left, and 2 to the power 20 is just over a million.

9

```
counts = {}
for c in ["5A", "6A", "5A"]:
    counts[c] = counts.get(c, 0) + 1
print(counts["5A"])
```

counts.get(c, 0) gives the count so far, or 0 the first time, and + 1 adds this row: it prints 2.

10 **An index makes searches faster, but it takes space and makes every new row slower to write.**

Indexes trade space and slower writes for faster searches, so index the columns you search by.